

Training Variational Autoencoders for Population Synthesis in Public Health with Missing Data

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Abstract—The importance of social determinants of health in shaping equitable public health policies is gaining increasing recognition. Emerging data sources, such as mobility and social media data, are becoming key to public health models. However, privacy concerns often limit access to these sensitive data, as even anonymized datasets are vulnerable to deductive disclosure. A promising solution to this challenge is the use of synthetic populations, which not only safeguard privacy but also enable the exploration of various “what-if” scenarios. Most existing studies on synthetic populations assume full access to complete datasets for training. However, in many public health applications, such as census data, this assumption is unrealistic due to incomplete responses, especially regarding sensitive questions like wealth or education. This paper introduces a novel approach for training Variational AutoEncoders (VAEs) with incomplete data, without resorting to missing value imputation. Instead, the VAEs are trained solely on the observed data. Using the 2019 PUMS dataset for Florida, we successfully train VAEs to generate diverse and flexible synthetic populations. By comparing marginal distributions and utilizing t-SNE for analysis, the results highlight the effectiveness of this method in addressing missing data challenges. This work demonstrates the potential of VAEs in generating synthetic populations for health research, even when complete datasets are unavailable, thereby offering a robust solution to advance public health studies while preserving data privacy.

1. Introduction

There is growing recognition of the crucial role that social determinants of health play in shaping equitable public health policies [1]. These determinants are increasingly captured in new data streams, such as mobility and social media data, which are becoming integral to public health models. However, broad access to this sensitive data is limited by privacy concerns; even without Personally Identifiable Information (PII), such data can still be vulnerable to deductive disclosure. A promising solution to this privacy issue is the use of synthetic populations trained on such data. Synthetic populations are artificially constructed representations of individuals that reflect the key distributions found in real populations, including characteristics like demographics, ac-

tivities, and health conditions. These models are gaining traction in health-related research and policy planning [2]. Beyond protecting privacy, synthetic populations have the added benefit of enabling the exploration of various “what-if” scenarios. However, conventional methods for generating synthetic populations face significant challenges: (i) they often fail to accurately represent smaller demographic groups, leading to skewed outcomes for underserved populations, and (ii) they lack computational efficiency, especially when adjusting to population changes (e.g., migration) or accounting for data uncertainty.

Traditionally, techniques like Synthetic Reconstruction (SR) and Iterative Proportional Fitting (IPF) have been used to create synthetic populations. For instance, Burger et al. [3] used these methods to model large-scale urban populations and social networks for agent-based simulations. While these approaches have been effective, they rely heavily on known marginal distributions and aggregate statistics, and they may not capture the intricate interdependencies and diversity within population attributes. Generative Adversarial Networks (GANs) offer a more advanced alternative that addresses many of the limitations of traditional methods. GANs are particularly well-suited for generating synthetic data in cases where data is scarce or sensitive. Unlike conventional techniques, GANs can produce high-quality synthetic data from limited samples, preserving the statistical integrity of the original dataset. This makes GANs especially useful for public health applications, where certain data—such as those representing minority groups—may be limited. GANs have demonstrated success in learning the probability distribution of complex, high-dimensional populations in an unsupervised manner [10]. Compared to SR and IPF, GANs are better equipped to model the nuanced interdependencies within real-world population data. This capability is crucial for accurately capturing the dynamics of social determinants of health, resulting in synthetic populations that more closely reflect actual demographic patterns. By simulating these complex interactions with greater realism, GAN-based models significantly reduce the biases and errors often found in traditional approaches.

A significant challenge that remains with GANs is their inability to learn the probability distribution of populations from datasets containing missing values. Current approaches

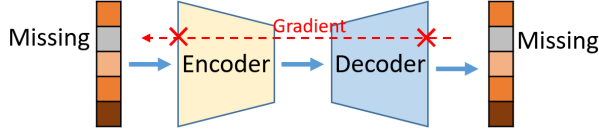


Figure 1. The overall architecture of the algorithm.

typically require an additional step of data imputation before training GANs or any other generative model. This process increases the training time since a separate model must be trained for imputing the missing data, followed by training the generative model. Moreover, training a data imputation model requires access to a complete dataset without missing values, which may not always be available. In this paper, we propose a method for training a Variational AutoEncoder (VAE) directly on datasets with missing values, eliminating the need for imputation. Our approach enables the creation of synthetic populations that better reflect demographic diversity. By training VAEs with missing data, we can generate synthetic data that closely mirrors real-world scenarios while addressing privacy concerns present in traditional datasets. This method strikes an important balance between privacy and data utility, offering a secure and efficient solution for generating synthetic populations.

2. Methodology

In this paper, we focus exclusively on Variational Autoencoders (VAEs) built with fully connected (FC) layers, as opposed to those using convolutional layers. This choice offers several advantages: 1) FC layers allow for independent training of each unit (node), meaning that units receiving missing values can be omitted during the training process. 2) FC layers are well-suited for capturing the latent representation of mixed categorical and numerical variables [10]. Let $\{x_n\}_{n=1}^N$ represent a set of N training samples, where $x_n \in \mathbb{R}^D$ denotes the input sample. The encoder $f_\theta(\cdot)$ and decoder $g_\phi(\cdot)$, both parameterized by unknown parameters θ and ϕ , transform the input x_n into its latent representation $h_n = f_\theta(x_n)$ and then back into the reconstructed input $x_n = g_\phi(h_n)$. Table 1 outlines the architecture of the encoder in our VAE model, with the decoder mirroring this structure.

TABLE 1. ENCODER CONFIGURATIONS FOR 9 PUMS FEATURES

Layer number	Layer Type	Configuration Parameter
1	FC	9×128
2	Leaky ReLU	-
3	Dropout	0.1
4	Batch Norm	128
5	FC	128×256
6	Leaky ReLU	-
7	Dropout	0.1
8	Batch Norm	256
9	FC	256×600

We make two key assumptions in our approach: 1) the missing values are distributed across all D attributes of the

input training vectors. For instance, if the first element of an input vector $x_n \in \mathbb{R}^D$ is missing, it is assumed that the first element is present in the remaining samples. 2) There are sufficient observations for each attribute within the training samples. While we do not perform imputation of missing values, it is essential that the observed attributes provide enough information for training. Additionally, it is important to highlight that only the original training data $\{x_n\}_{n=1}^N$ contain missing values, and missing values do not propagate into the hidden layers. Therefore, training fully connected layers with missing attributes applies solely to the initial layers of the encoder and the final layer of the decoder.

Let $\mathbf{W} = [\mathbf{w}_1 | \mathbf{w}_2 | \dots | \mathbf{w}_D]$ denote the weight matrix of a fully connected (FC) linear layer, consisting of D columns, each with K elements, where $\mathbf{w}_d \in \mathbb{R}^K$, resulting in $\mathbf{W} \in \mathbb{R}^{K \times D}$. The linear layer maps an input vector $\mathbf{x} \in \mathbb{R}^D$ to an output vector $\mathbf{h} \in \mathbb{R}^K$ using the relation $\mathbf{h} = \mathbf{W}\mathbf{x} + \mathbf{b}$, where $\mathbf{x} = [x_1, x_2, \dots, x_D]^T$, $\mathbf{h} = [h_1, h_2, \dots, h_K]^T$, and $\mathbf{b} \in \mathbb{R}^K$ represents the bias, capturing the residuals that cannot be modeled by the linear term $\mathbf{W}\mathbf{x}$, i.e., $\mathbf{b} = \mathbf{h} - \mathbf{W}\mathbf{x}$. Therefore, each output element h_k can be written as: $h_k = \sum_{d=1}^D x_d w_{kd} + b_k$ $k = 1, 2, \dots, K$. Equation $\mathbf{h} = \mathbf{W}\mathbf{x} + \mathbf{b}$ signifies the output of an FC layer as a linear combination of vectors $\{\mathbf{w}_d\}_{d=1}^D$, where the input $\mathbf{x}$ serves as the coefficients for this linear combination. The vectors $\mathbf{w}_d$ are estimated from the training samples by solving the optimization problem: $\sum_{d=1}^D \sum_{n=1}^N (h_{n,d} - \mathbf{w}_d x_{n,d} - b_d)^2$. To facilitate training with missing values, we adopt a strategy where $\mathbf{w}_d$ responsible for processing missing values are temporarily excluded. This entails removing the units of the FC layer that receive missing values, along with all associated outgoing connections.

3. Experimental Results

In this section, we outline the experimental procedure chosen to validate our proposed method.

3.1. Data

The Public-Use Microdata Sample (PUMS) data utilized in this study was obtained from the United States Census Bureau [6] and curated by selecting nine key attributes: PINCP (Total person's income), AGE (Age), QTRBIR (Quarter of birth), POVPI (Income-to-poverty ratio), ANCI (Recoded Detailed Ancestry - first entry), POBP (Place of birth), HINS1 (Health insurance through a current or former employer or union), SEX (Sex), and SCHL (Educational attainment). The 2019 PUMS dataset for Florida originally comprised 205,296 samples. These attributes were specifically chosen for their relatively even distributions to minimize the risk of mode collapse [7].

After narrowing down the attributes, rows with missing data were excluded from the dataset. This refinement left 165,556 samples, representing the state of Florida. The

Comparative Histograms of Real vs 20% Missing Synthetic

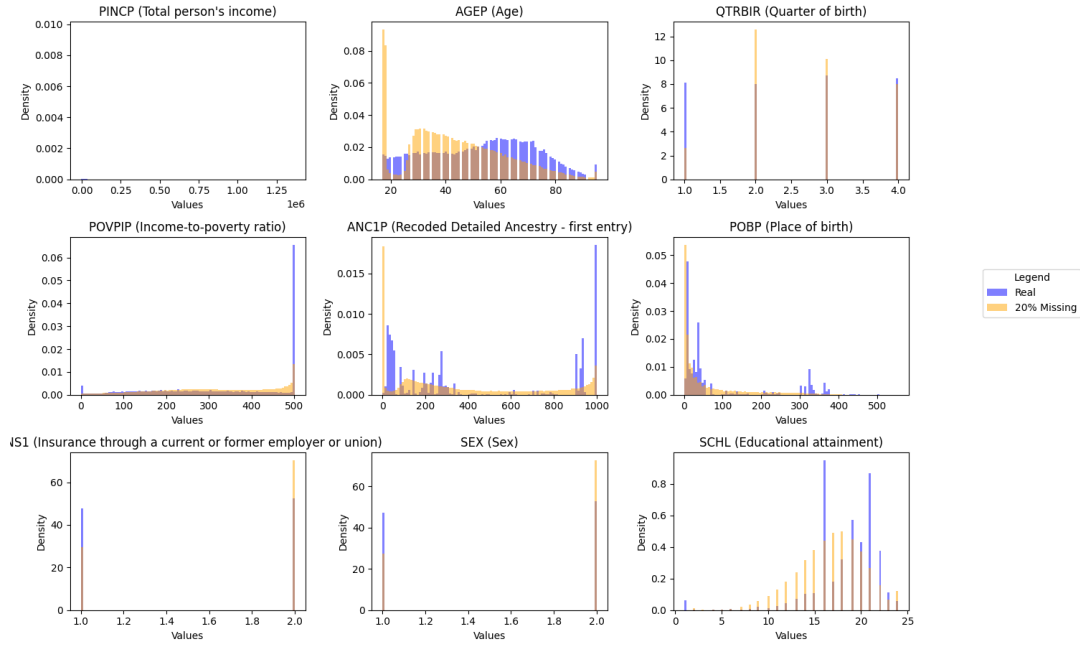


Figure 2. The histogram of the real and synthetic data when the VAE is trained with dataset having 20% missing values for all nine attributes.

selected attributes vary significantly in their ranges—for example, AGEP (Age) spans from 17 to 95 years, while PINCP (Total person’s income) ranges from approximately −6,800 to 1,367,100. To address these inconsistencies, each attribute was normalized to a scale between 0 and 1.

In our previous work [10], we trained our models on a complete dataset by removing rows with missing values. However, in this study, we introduce a method that enables training the network with missing data, without the need for imputation. To illustrate this, we simulate missing values by randomly removing data points during training while still successfully training the network. Suppose the training input sample $x_n \in \mathbb{R}^D$ is drawn from the set $\{x_n\}_{n=1}^N$. We randomly select k attributes out of the total D attributes in the input sample $x_n \in \mathbb{R}^D$ and set them to missing, resulting in a percentage of missing values equal to $100 * \frac{k}{D}$.

3.2. Experiment

Validating synthetic population generation from tabular datasets presents distinct challenges. Existing quantitative metrics for evaluating generative models are often insufficient for tabular datasets that lack clear class distinctions. Metrics like Inception Score (IS) and Fréchet Inception Distance (FID) rely on Kullback–Leibler (KL) divergence [9], which requires both real and synthetic data distributions. In computer vision applications, synthetic data is classified using a pretrained model (e.g., VGG16, ResNet50) on a separate dataset, and the resulting class distributions are used to compute KL divergence, generating IS and FID

scores. However, due to the absence of well-defined classes in tabular data, histograms of the marginal distributions for real and synthetic data are presented in Fig. 2 as a validation method. The figure illustrates the marginal distribution of each variable in both real and synthesized data, with the VAE trained on a dataset where 50% of the features per sample are missing. The synthesized data’s marginal distributions closely match those of the real dataset, consistent with prior findings [10]. This alignment underscores the effectiveness of the proposed method in learning the complete data distribution from partially observed data, i.e., training data with missing values.

To further evaluate the effectiveness of our proposed approach, we conducted an ablation study examining the impact of different percentages of missing values per training sample. In the first experiment, no values were removed (i.e., 0% missing values), and the VAE was trained to learn the data distribution. In the second experiment, we randomly removed 20% of the features per sample and trained the VAE. We then synthesized samples from both VAEs and compared their marginal distributions, as shown in Fig.3. The figure demonstrates that the marginal distributions of the synthetic datasets in both cases are similar. In the third experiment, we removed 50% of the features per sample and trained a VAE with these missing values. The resulting synthetic samples are then compared to those generated by the VAE trained with 0% missing values, using their marginal distributions (Fig.4). Some features, such as ANC1P in Fig. 2, show slight differences between the marginal distributions of the synthetic and real data. Moreover, as the percentage of the

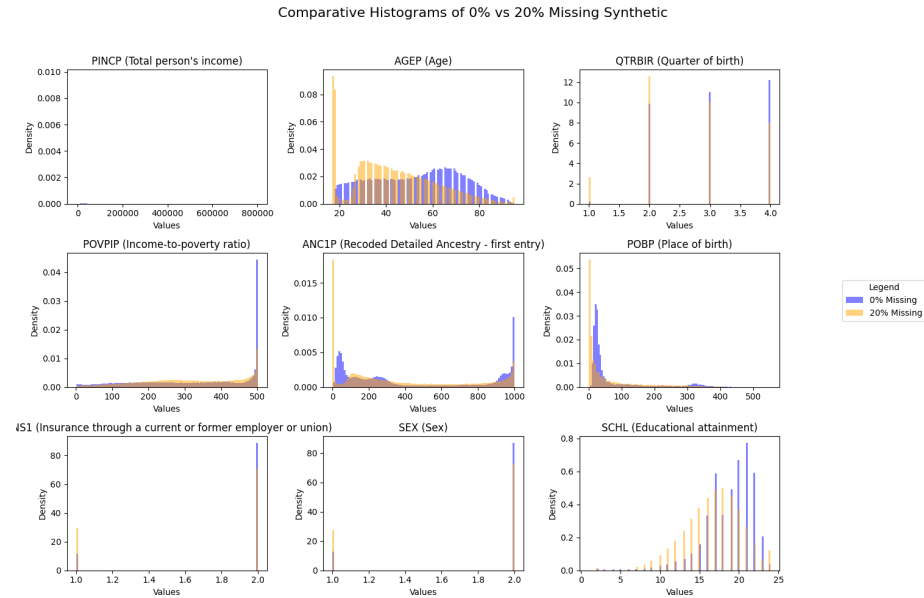


Figure 3. The histogram of the synthetic data when the VAE is trained with dataset having 0% and 20% missing values for all nine attributes.

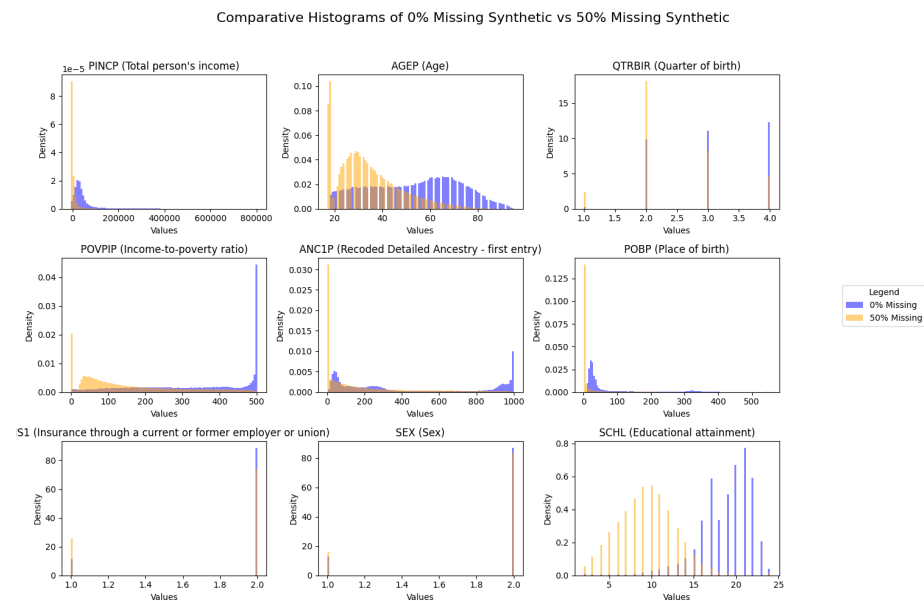


Figure 4. The histogram of the synthetic data when the VAE is trained with dataset having 0% and 50% missing values for all nine attributes.

missing value is increases in Fig. 4, the distribution of the synthetic data slightly deviates from the real data. This phenomenon is more visible in the variables SCHL and AGE.

4. Conclusion

This study highlights the potential of Variational Autoencoders (VAEs) to generate synthetic populations for public health applications. Our methodology successfully addresses a key challenge—training VAEs on datasets with

missing values without the need for imputation. While this paper introduces a novel and powerful tool for public health research, offering promising opportunities for data-driven policymaking and analysis, further development is still needed. Future work will concentrate on refining the mathematical framework for training VAEs with missing data and tackling the issue of mode collapse within VAEs.

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